

Homework 5

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Exercise 1

Let there be n firms. Suppose the inverse demand function $P(Q)$ is:

$$P(Q) = \begin{cases} a - Q & Q \leq a \\ 0 & Q > a \end{cases}$$

and each firm has the cost function $c_i(q_i) = cq_i^2$, with $c < a$. Letting $q_{-i} = Q - q_i$, the profit function $\pi_i(q_i, q_{-i})$ is:

$$\begin{aligned} \pi_i(q_i, q_{-i}) &= P(Q)q_i - cq_i^2 \\ &= \max(a - q_i - q_{-i}, 0)q_i - cq_i^2 \end{aligned}$$

Let us assume $a - q_i - q_{-i} \geq 0$. Taking the first derivative with respect to q_i :

$$\begin{aligned} \frac{\partial \pi_i}{\partial q_i} &= 0 = (a - q_{-i}) - 2(1 + c)q_i \\ q_i &= \frac{a - q_{-i}}{2(1 + c)} \end{aligned}$$

Now taking the second derivative:

$$\frac{\partial^2 \pi_i}{\partial q_i^2} = -2(1 + c)q_i < 0$$

The best response is then indeed $q_i^* = \frac{a - q_{-i}}{2(1 + c)}$, making $(q_1^*, \dots, q_n^*)$ a Nash equilibrium.

Suppose firms i and j have best responses q_i^* and q_j^* , respectively, with $q_i^* \neq q_j^*$ and let $Q'^* = Q^* - q_i^* - q_j^*$. Then:

$$2(1 + c)q_i^* = a - Q'^* - q_j^* 2(1 + c)q_j^* = a - Q'^* - q_i^*$$

It then follows that:

$$\begin{aligned} 2(1 + c)(q_i^* - q_j^*) &= q_i^* - q_j^* \\ (1 + 2c)(q_i^* - q_j^*) &= 0 \end{aligned}$$

For this to be true, $q_i^* = q_j^*$. This is a contradiction; therefore in a Nash equilibrium, all firms produce the same output q^* . That output is:

$$q^* = \frac{a - (n-1)q^*}{2(1+c)}$$

$$q^* = \frac{a}{1+2c+n}$$

The equilibrium price is:

$$P^* = a - Q^* = a - n \cdot \frac{a}{1+2c+n}$$

$$= \frac{a(1+2c)}{1+2c+n}$$

We see that $\lim_{n \rightarrow \infty} P^* = 0$, so the price decreases as the number of firms increases.